

Achraf Chakroun

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<https://muffinxz.github.io/cv-web/>

PROFILE

I hold a Master's degree in Engineering from the University of Québec at Rimouski (UQAR), graduating with 4.1/4.3 GPA and I'm also a graduate of the National Engineering School of Tunis (ENIT). My work sits at the intersection of applied AI engineering and reliability: I build and improve agentic LLM systems with a focus on reduced hallucinations, measurable performance, and production guardrails. I'm also interested in hardware optimization of computer vision models by making the deployed edge AI losing latency and maintaining accuracy. I am also fluent in English and French.

KEY SKILLS

• AI / ML & Agentic Systems:

• LLM applications, RAG, prompt engineering, tool use, agent orchestration (LangChain, LangGraph), Deep Learning, Computer Vision, PyTorch, TensorFlow, Hugging face

• Databases / Testing / Deployment:

• PostgreSQL, SQLAlchemy, Alembic, Pytest, Render, Docker, Linux, Git, FastAPI

• Hardware Acceleration / Edge AI:

• FPGA & toolchain: VHDL, Vivado, Vitis HLS, Vitis AI, Xilinx System Generator (XSG)

• Edge optimization: PTQ/QAT quantization, pruning (static/dynamic), hardware-aware deployment

• Scientific / Engineering Tools:

• MATLAB, Signal and image processing

• **Collaboration:** Jira, Confluence, Microsoft Teams, Trello, Slack

PROFESSIONAL EXPERIENCE

Generative AI Engineer (Contract)

Sep. 2025 – Feb. 2026

Opal-RT Technologies

- Built an engineering-grade AI assistant converting natural-language requirements into versioned JSON component specs and standards-compliant C modules for HYPERSIM DLL integration, cutting time-to-first-working component from ~4 days to ~2 minutes while keeping validation + acceptance checks in the loop.
- Designed a LangGraph orchestration (intent routing → spec diff/update → validation → codegen) with auditable state transitions, enabling repeatable runs, rollback, and consistent outputs across sessions.
- Improved grounding using RAG over IEEE/CIGRE documentation + internal reference components, adding citation-driven generation and edge-case retrieval to reduce unsupported assumptions and improve consistency on tricky specs.
- Hardened production behavior with deterministic codegen fallbacks, JSON schema/Pydantic validation, compilation/format checks, and automated acceptance tests, preventing "plausible-but-wrong" code and catching failures early.
- Worked in a GitLab feature-branch + merge-request workflow, using local pytest + Ruff checks to keep changes reviewable and reliable.

Research assistant — Hardware Implementation of YOLOv5 on Kria KV260 FPGA

Sep. 2022 – Sep. 2025

University of Quebec at Rimouski (UQAR)

Project: Hardware acceleration of the object detection model YOLOv5 using FPGA board via Vitis-AI, achieving 15.78x less power consumption on FPGA (2.7W) compared to a GPU (42.7W) with same overall detection performance (7.03 ms/142 fps) as the host machine with RTX 3070ti (6.33 ms/158 fps).

- Conducted extensive bibliographic research on object detection architectures (Backbone, Neck, Head), acceleration techniques (Quantization, Pruning) and hardware acceleration platforms (GPU/FPGA).
- Analyzed the YOLO model in detail, particularly YOLOv5, and studied the specific features of the Kria KV260 FPGA board.
- Trained 3 versions of YOLOv5 (nano, small and large) on 3 different applications using YAML files : Skin cancer detection (using the HAM10000 dataset), autonomous driving (using KITTI dataset) and outdoor environment scanning (using COCO2017 dataset).
- Quantized the models with post training quantization (PTQ) and optimized the YOLOv5 model, using Vitis-AI with the PyTorch framework.
- Profiled the model using netron on host computer and Vitis-AI profiler on the FPGA board.
- Developed an executable for image/video inference and real-time inference using OpenCV/GStreamer and Cmake commands.

Teaching Assistant – Logic Circuits for undergraduate students**Jan. 2025 – Apr. 2025****University of Quebec at Rimouski (UQAR)**

- Supervised the mastery of number systems (binary, octal, hexadecimal) and coding systems (ASCII, natural binary, reflected binary).
- Guided the design and simplification of combinational logic circuits (Boolean algebra, De Morgan's theorems, Karnaugh method).
- Supervised the design of sequential circuits (flip-flops, registers, counters, memories, finite state machines).
- Implemented and validated digital circuits on FPGA with an introduction to HDL (hardware description language).
- Corrected and evaluated scientific reports on practical work.

Teaching Assistant – Signal Processing for undergraduate level**Sep. 2024 – Dec. 2024****University of Quebec at Rimouski (UQAR)**

- Supervised students in signal sampling and quantification.
- Guided frequency analysis and the application of Z-transforms.
- Taught amplitude modulation and demodulation.
- Corrected and evaluated scientific reports on practical work.

Teaching Assistant – Digital Systems Prototyping for graduate students**Jan. 2024 – Aug. 2024****University of Quebec at Rimouski (UQAR)**

- Supervised the design of digital systems using VHDL.
- Introduced synthesis methods based on high-level languages for mixed systems.
- Assisted in the use of software/hardware simulation tools (Vivado, Vitis).
- Supervised the programming of FPGA boards (Basys-3, Nexys-4, PYNQ).
- Corrected and evaluated scientific reports on practical work.

End-of-studies internship**Jun. 2020 – May 2021****Laboratory of Robotics, Computing, and Complex Systems (RISC-ENIT)**

- Designed and implemented spatio-spectral classification techniques for hyperspectral images applied to agriculture.
- Synthesized hyperspectral images using the USGS spectral library.
- Developed and trained convolutional neural networks for hyperspectral image classification.

PROJECTS

Todo Web Application (<https://prot-v0xc.onrender.com>)**FastAPI, PostgreSQL, SQLAlchemy, Alembic, Jinja2, JavaScript, Pytest, Render**

- Built and deployed a full-stack task management web application using FastAPI, PostgreSQL, SQLAlchemy, Jinja2, and JavaScript.
- Implemented user authentication with registration, login, password hashing using bcrypt, and JWT-based access control for protected routes.
- Developed todo CRUD functionality, integrated frontend forms with backend APIs, and validated request handling with Pydantic models.
- Managed database schema evolution with Alembic and wrote Pytest test cases to validate application behavior and improve reliability.
- Deployed the application on Render and debugged production issues involving database configuration, template resolution, routing, and API validation.

AI Calorie Tracker — Multimodal LLM Vision App**May 2026**

- Built a multimodal AI prototype that analyzes food images and generates calorie estimates using Ollama llama3.2-vision.
- Implemented image preprocessing with PIL and base64 encoding to send image payloads to a local vision-language model.
- Designed structured prompts to improve the quality and consistency of nutrition analysis responses.
- Developed a lightweight Gradio interface for interactive image upload, prompt editing, and real-time model output display.
- Demonstrated practical integration of computer vision, LLM prompting, local inference, and user-facing AI prototyping.

EDUCATION

Master of Engineering (Research) University of Quebec at Rimouski (UQAR)	Sep. 2022 – Dec. 2025
National Engineering School of Tunis Bachelor's degree in Electrical Engineering	Sep. 2017 – Jul. 2021
Preparatory diploma (Mathematics - Physics) Preparatory Institute for Engineering Schools in Tunis	Sep. 2014 – Jul. 2017

CERTIFICATIONS

FastAPI - The complete course 2026	Udemy
HYPERSIM® - Power Systems Real-Time Simulation	Opal-RT Technologies
TEF Canada (Test d'Évaluation de Français pour le Canada)	Le Français des affaires
Developing AI Inference Solutions with the Vitis AI Platform	AMD/Xilinx
TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning	Coursera and Laurence Moroney, Google
International English Language Testing System (IELTS 8.5/9)	IELTS – British Council

AWARDS AND PRIZES

Jacques Dionne Excellence Scholarship (Recognition of outstanding academic achievement)	March 2024
Participation in the FI3E Innovation Forum (Academic and industrial event focusing on technological innovation)	April 2025

REFERENCES

Vincent Lapointe , Chef d'Équipe - Innovation / Team lead - Innovation vincent.lapointe@opal-rt.com	Opal-RT Technologies
Mohammed Bahoura , Ing., Ph.D. mohammed_bahoura@uqar.ca	University of Quebec at Rimouski (UQAR)